

Nitrogen starvation induces extensive changes in the redox proteome of *Prochlorococcus* sp. strain SS120

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Summary

Very low nitrogen concentration is a critical limitation in the oligotrophic oceans inhabited by the cyanobacterium *Prochlorococcus*, one of the main primary producers on Earth. It is well known that nitrogen starvation affects redox homeostasis in cells. We have studied the effect of nitrogen starvation on the thiol redox proteome in the *Prochlorococcus* sp. SS120 strain, by using shotgun proteomic techniques to map the cysteine modified in each case and to quantify the ratio of reversibly oxidized/reduced species. We identified a number of proteins showing modified cysteines only under either control or N-starvation, including isocitrate dehydrogenase and ribulose phosphate 3-epimerase. We detected other key enzymes, such as glutamine synthetase, transporters and transaminases, showing that nitrogen-related pathways were deeply affected by nitrogen starvation. Reversibly oxidized cysteines were also detected in proteins of other important metabolic pathways, such as photosynthesis, phosphorus metabolism, ATP synthesis and nucleic acids metabolism. Our results demonstrate a wide effect of nitrogen limitation on the redox status of the *Prochlorococcus* proteome, suggesting that besides previously reported transcriptional changes, this cyanobacterium responds with post-translational redox changes to the lack of nitrogen in its environment.

Introduction

Prochlorococcus is a cyanobacterium (Chisholm *et al.*, 1988) which has become a major model in marine ecology (Coleman and Chisholm, 2007) for different

reasons. Since it is the most abundant photosynthetic organism in our planet (Partensky and Garczarek, 2010), it provides a significant contribution to primary production on Earth (Goericke and Welschmeyer, 1993; Liu *et al.*, 1997). It has a number of features which have attracted the interest from scientists, namely the small size of its genome (Dufresne *et al.*, 2003; Rocap *et al.*, 2003), the existence of a large collection of cultured isolates (Moore *et al.*, 2007) – representatives of different environments in the ocean – and the availability of a growing number of *Prochlorococcus* genomes (Coleman *et al.*, 2006; Kettler *et al.*, 2007).

One of the most distinguishing features of this cyanobacterium is its outstanding ability to grow at very high concentrations in oligotrophic regions of the intertropical oceans (Liu *et al.*, 1997; Johnson *et al.*, 2006). In such areas permanently subjected to nutrient limitations (Howarth, 1988), *Prochlorococcus* outgrows its closer competitor, *Synechococcus*, keeping very high concentrations down to almost 200 m depth (Partensky *et al.*, 1999a). This remarkable fact has been attributed to a number of different causes, such as their high efficiency in nutrient scavenging due to its high surface/volume ratio (Partensky *et al.*, 1999b). Yet if we consider the tremendous amount of biomass that is comprised by *Prochlorococcus* cells in such large regions, it is obvious that this cyanobacterium had to evolve strategies to cope with nutrient limitations in order to keep its metabolism as unaffected as possible. Some examples of such adaptations have been reported in literature as the utilization of sulfolipids instead of phospholipids to reduce the phosphorus needs in phytoplankton (Van Mooy *et al.*, 2006). The utilization of Pcb as antenna proteins in *Prochlorococcus* evolved from stress-induced proteins (*isiA*-like), instead of the bulkier phycobilisomes (Partensky and Garczarek, 2003) and the reduction in nitrogen content in the amino acid sequences (Grzymalski and Dussaq, 2012). However, the severe oligotrophy observed in the habitats where *Prochlorococcus* is more successful suggests that this organism has been subjected to nutrient limitation [and in particular, nitrogen limitation (Graziano *et al.*, 1996)] for a very prolonged period of time, leading most possibly to a wide effect on different aspects of cell metabolism. Nitrogen is one of the key limiting nutrients in oligotrophic oceans, reaching levels in the nanomolar range (Capone, 2000). Different regions in the oceans

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have been described to be permanently limited in this element (Zehr and Kudela, 2011).

Light-dependent disulfide exchange on cysteine residues of proteins, catalysed by the Thioredoxin (Trx) reductase, was first identified in chloroplasts and since then redox modifications and subsequent redox regulation of proteins have been linked to photosynthesis, storage, transcription, translation and protein folding (Buchanan and Balmer, 2005). The reversible modification of cysteine residues in proteins, such as oxidation, nitrosylation and formation of disulfide bonds, is a method widely utilized by cells to control protein structure and function and offers a means of reversibly regulating the activity of a significant number of important enzymes. The redox status of the thiol group is subjected to reversible reactions, allowing rapid and effective mechanisms to control enzyme activity. In previous studies, it has been shown how the redox proteome of different organisms is profoundly affected by different environmental conditions. The disulfide redox proteome of the cyanobacteria *Synechocystis* sp. PCC 6803 has been extensively studied by Lindahl and Florencio (Florencio *et al.*, 2006; Perez-Perez *et al.*, 2006; Mata-Cabana *et al.*, 2007), who identified almost 80 potential Trx protein targets, with a high overlap with those identified as Trx targets in the chloroplasts of higher plants (Lindahl and Kieselbach, 2009). Our group has demonstrated that strong decreases in the N/C balance induce different changes in key enzymes, such as glutamine synthetase (GS) and isocitrate dehydrogenase (IDH), including both the expression of the corresponding genes (López-Lozano *et al.*, 2009) and the abundance of the proteins (El Alaoui *et al.*, 2001; 2003; López-Lozano *et al.*, 2009). Furthermore, different experiments suggest that both enzymes are subjected to regulation by redox oxidative modifications (Gómez-Baena *et al.*, 2001; 2006). This led us to explore the possible involvement of redox proteome changes in the response of *Prochlorococcus* to key environmental conditions, such as nitrogen starvation, which is believed to be one of the most important limitations for the growth of phytoplankton in oligotrophic oceans. In this work, we utilized novel techniques to detect changes in cysteine-containing proteins affected by nitrogen starvation, finding variations in several proteins both directly involved in nitrogen metabolism and other pathways that may be due to secondary adaptive effects.

Effects of nitrogen starvation on protein glutathionylation in *Prochlorococcus*

The object of this work was to address the effects of nitrogen starvation on the *Prochlorococcus* redox proteome. Previous studies carried out by our group demonstrated that nitrogen starvation had little effect on a key enzyme in nitrogen metabolism, GS, which showed no

significant changes either in the activity or concentration (El Alaoui *et al.*, 2001; 2003). However, in studies focused on metal-catalysed oxidative systems, we observed that GS and also IDH from *Prochlorococcus* were subjected to a process of oxidative inactivation (Gómez-Baena *et al.*, 2001; 2006). These results suggested that oxidative mechanisms might be involved in the regulation of metabolism in *Prochlorococcus*. This prompted us to study whether one of the most important limitations suffered by natural populations of *Prochlorococcus* – i.e. very low concentrations of nitrogen, given that oligotrophic oceans is the preferred habitat of this marine cyanobacterium – might produce metabolic effects through redox signalling. In order to check this hypothesis, we carried out Western blotting of control vs starved samples, to detect glutathionylated proteins. The results are shown in Fig. 1, comparing control samples of *Prochlorococcus* sp. strain SS120 with samples of cultures subjected to nitrogen starvation for 24 h, mimicking the conditions of limited nitrogen availability in the intertropical oceans. A large increase in the protein glutathionylation was observed in all starved samples, suggesting that lack of nitrogen had a strong effect on the redox status of the *Prochlorococcus* sp. strain SS120 proteome. Consequently, we decided to further explore the effects of nitrogen limitation in particular, since nitrogen has been reported to be limiting in very wide regions of the oceans.

Effect of nitrogen starvation on the *Prochlorococcus* redox proteome

Prochlorococcus SS120 cultures growing on standard, ammonium-containing PCR-S11 medium were transferred to medium without added nitrogen, as explained previously (El Alaoui *et al.*, 2003). Cells were harvested after 24 h and processed to detect reversible cysteine modifications by the shotgun redox proteomics method (McDonagh *et al.*, 2009). Three independent biological replicates were used for both control and N-starved cells. This method not only identifies the proteins involved but also the precise cysteine within each protein that is subject to a reversible modification. This has a number of advantages over traditional shotgun proteomic techniques which can be prone to undersampling and can sometimes be overwhelmed by peptides that ionize and fragment well in MS/MS experiments, and for this reason there can be little overlap between MS runs (Malmstrom *et al.*, 2007; Nilsson *et al.*, 2010). As we select for only those peptides containing reversibly oxidized cysteines, we are greatly reducing the number of peptides and increasing confidence that they are in fact redox sensitive. However, some caution should be used in the interpretation of results, and if the peptide is not detected the cysteine residue might be either reduced or overoxidized

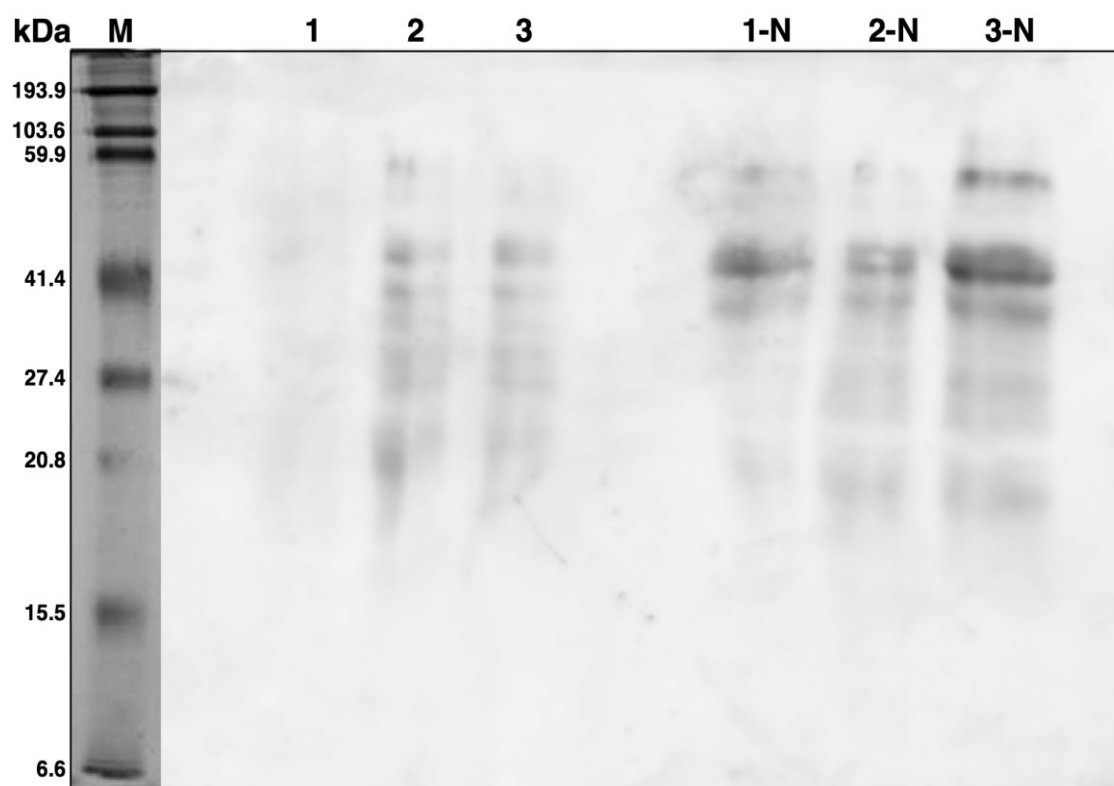


Fig. 1. Immunoblot detection of glutathionylated proteins in *Prochlorococcus* SS120 cells subjected to nitrogen starvation. Three independent cultures were divided in two aliquots each, and subjected to control conditions (lanes 1, 2 and 3) or to nitrogen starvation for 24 h (lanes 1-N, 2-N and 3-N). Glutathionylated proteins were detected by immunostaining as described (McDonagh *et al.*, 2011b).

to irreversible sulfinic or sulfonic forms or alternatively the gene might not be expressed. Also, the cys-containing tryptic peptide must be amenable to analysis by MS.

Using the stringent identification criteria utilized in this study, 80 proteins were identified between control and nitrogen-starved groups (Table S1). Of these, 53 proteins were consistently identified in all biological replicates from both conditions. Four proteins were identified from peptides not containing cys residues and would be considered as false positives due to misidentification by the search engine or because the peptide is attached to the biotin-HPDP (N-[6-(Biotinamido)hexyl]-3'-(2'-pyridyldithio) propionamide). Grouping those 80 proteins into general clusters reveal that 20 were involved in nucleotide or amino

acid biosynthesis, 17 in energy or metabolic processes, 12 in transcription and translation, 10 forming part of photosynthetic system, 2 in chaperone or stress response and the rest in signalling or unknown processes.

There were seven proteins (Table 1) that were only identified as containing peptides with reversibly modified cysteines in either the control or nitrogen-starved group. Proteins only identified in the control group were ABC transporter, substrate binding protein (cys67), formyltetrahydrofolate deformylase (cys13), pyruvate dehydrogenase E1 component, alpha subunit (cys277), ribose-phosphate pyrophosphokinase (cys247) and UDP-3-O-(3-hydroxymyristoyl) glucosamine N-acyltransferase (cys98). Two proteins were identified with reversibly oxi-

Table 1. Proteins with reversibly oxidized cysteines detected only under either control or N starvation conditions.

Protein	GI number	Cys peptide detected	Condition detected
ABC transporter, substrate binding protein, possibly Mn	157413499	67	Control
Formyltetrahydrofolate deformylase	78780180	18	Control
Pyruvate dehydrogenase E1 component, alpha subunit	254525916	271	Control
Ribose-phosphate pyrophosphokinase	33861636	247	Control
UDP-3-O-(3-hydroxymyristoyl) glucosamine N-acyltransferase	91070309	98	Control
Isocitrate dehydrogenase	78780073	463	N starvation
Ribulose-phosphate 3-epimerase	33861323	68	N starvation

Table 2. Proteins containing reversibly oxidized cysteines with significant differences in oxidation state between N-starved and control conditions.

Protein	GI number	Cys detected	Fold change	P-value
Bifunctional 3,4-dihydroxy-2-butanone 4-phosphate synthase	72382157	Cys 43/65	1.7	0.05
Chaperonin GroEL	123969170	Cys146	2.0	0.064
Coproporphyrinogen III oxidase	123969321	Cys231	1.7	0.051
DNA-directed RNA polymerase subunit beta	124022146	Cys886/875	1.9	0.003
F0F1 ATP synthase subunit beta	159904098	Cys227	3.1	0.043
Flavodoxin FldA	126696711	Cys142	1.6	0.028
Fructose 1,6-bisphosphatase II	126696165	Cys149	2.1	0.006
Homoserine dehydrogenase	78779445	Cys344	5.8	0.045
(3R)-hydroxymyristoyl-ACP dehydratase	123969068	Cys117	1.8	0.062
*L-asparaginase II	123966129	Cys166	0.3	0.004
Phosphoribosylformylglycinamide synthase	33861337	Cys89	1.8	0.011
Photosystem I subunit VII	159904273	Cys21/34	1.9	0.024
Photosystem II reaction centre D1	33239704	Cys19	1.7	0.033
Preprotein translocase subunit SecA	126697168	Cys577	1.5	0.086
Putative inorganic pyrophosphatase	123968086	Cys163	1.7	0.041
Putative O-acetyl homoserine sulfhydrylase	33861199	Cys96	2.0	0.04
Putative urea ABC transporter, substrate binding protein	78779213	Cys145	5.3	0.021
RNA recognition motif-containing protein	123967549	Cys30	1.8	0.06
Triosephosphate isomerase	123968574	Cys127	1.9	0.03

L-asparaginase II is marked with an asterisk since it is the only case showing a decrease in this table.

dized cysteines only in nitrogen-starved cells: IDH (cys463) and ribulose phosphate 3-epimerase (cys68). This points to a direct response of the cell to nitrogen limitation and indeed our group has previously identified three of these yeast protein orthologues as being redox sensitive (Idh2p, Rpe1p and Prs3p) (McDonagh *et al.*, 2011a). The two latter proteins have been described as thioredoxin targets in *Synechocystis* PCC 6803 (Lindahl and Kieselbach, 2009).

As a further analysis we used the Progenesis LC-MS program to relatively quantify the abundance of detected peptides between test groups. Using an ANOVA analysis, 10 of the proteins detected in all samples (Table S1) showed a significant increase in detection of the reversibly oxidized cysteine containing peptide in nitrogen-starved cells (Table 2).

Interestingly, we found only two proteins directly related to nitrogen metabolism that showed significant differences between groups according to Progenesis LC-MS: the substrate binding protein of the urea ABC transporter (binding protein) and L-asparaginase II. It is worth noting that when looking at the whole list of detected proteins, some proteins in Table 2 did not meet the criteria for inclusion in Table S1, which may be due to processing of raw data in run alignments, filtering of data and file conversions between the different programs. The urea ABC transporter binding protein had a 5.3-fold increase in the N-starved samples and the peptide detected contained cys145. This corresponds to the only conserved cysteine residue found in this protein in cyanobacteria (not shown).

Enzymes comprise the majority of the proteins detected by this method, given that most of the proteins possessing cysteines subjected to reversible oxidation belong to that

category. When looking at the complete list of proteins (Table S1) detected by this method, we found two of the key enzymes in the nitrogen metabolism, GS and IDH. These enzymes represent respectively the main pathway for nitrogen assimilation (GS/GOGAT) and the link between nitrogen and carbon metabolisms (IDH). We also detected the peptide containing cys279 of the ferredoxin NADP oxidoreductase that can provide the reducing power for Trx and subsequently target proteins.

There were enzymes involved in the transfer of nitrogen groups – as amine or amide – and also peptidases. These include: glutamate-1-semialdehyde aminotransferase, glutamine amidotransferase class-I: CTP synthase, aspartyl-glutamyl-tRNA amidotransferase subunits A and B, porphobilinogen deaminase, glutamate-1-semialdehyde amidotransferase; leucyl aminopeptidase.

Effects of nitrogen starvation on nitrogen-related enzymes

Figure 2 shows a Western blotting of GS from the same samples utilized in the redox proteome experiments; nitrogen starvation induced a clear increase (c. threefold) in the concentration of this enzyme. Previous reports by Tolonen and coworkers (Tolonen *et al.*, 2006) showed also a significant increase in the expression of *glnA*, encoding GS in cyanobacteria. However, this response seems to depend on the strain, since our team has previously shown that GS concentration was clearly increased after 120 h of starvation in *Prochlorococcus* strain SS120, in good agreement with the results shown here, but not in the strain PCC 9511 (El Alaoui *et al.*, 2003).

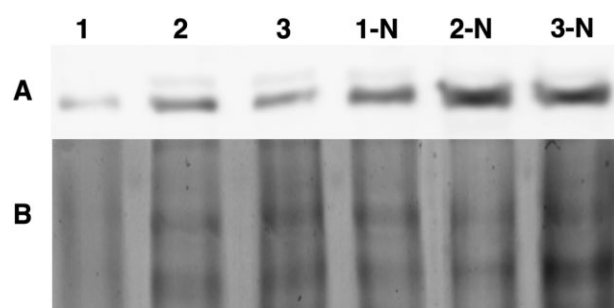


Fig. 2. Changes in glutamine synthetase concentration in *Prochlorococcus* SS120 cells subjected to nitrogen starvation. Cell extracts were subjected to Western blotting, and glutamine synthetase detected by using antibodies raised against glutamine synthetase from *Synechocystis* sp. PCC 6803 (El Alaoui *et al.*, 2003). Three independent cultures were divided in two aliquots each, and subjected to control conditions (lanes 1, 2 and 3) or to nitrogen starvation for 24 h (lanes 1-N, 2-N and 3-N). A, Western blotting. B, Coomassie-stained gel.

Figure 3 shows the effect of nitrogen starvation on the IDH concentration of the *Prochlorococcus* sp. SS120 strain. Despite the variability observed in controls, and in contrast to the results observed for GS, we observed no clear change under nitrogen limiting conditions, with respect to the control culture on standard culture medium. This result fits nicely with the fairly constant expression of *icd* in the *Prochlorococcus* sp. SS120 strain, the gene encoding for IDH, under nitrogen starvation reported previously (López-Lozano *et al.*, 2009) and could point to an effective post-translational redox control mechanism for the protein.

Discussion

The main goal of this study was to utilize an unbiased, shotgun-based proteomic technique to find out whether nitrogen starvation, a wide limitation in intertropical oceans, provokes changes in the redox status of the *Prochlorococcus* proteins possessing cysteines sensitive to reversible oxidation.

Our results demonstrate this is indeed the case, therefore suggesting that redox modifications could mediate the physiological response of this marine cyanobacterium to the lack of an important element as nitrogen. To our knowledge, this is the first proteomic study on the effects of nitrogen limitation in *Prochlorococcus*.

Table 2 shows changes in the peptides detected and quantified which are statistically significant and the fold change ranged from 1.7 to 2.0. The fact these peptides were identified in all samples would potentially point to a redox modification in many constitutively expressed proteins that could alter the metabolic flow through the respective pathways, offering a means for the cell to reversibly control the activity of certain key enzymes potentially affecting key signalling pathways. This sug-

gests that *Prochlorococcus* adapts to the lack of nitrogen by triggering proteomic changes.

A number of proteins were only detected under either control (i.e. Mn ABC transporter or formyltetrahydrofolate deformylase) or N-starved conditions (i.e. IDH or ribulose-phosphate 3-epimerase; Table 1). In addition, two of the cysteine-modified proteins were specifically oxidized under nitrogen starvation. This feature reinforces the importance of the observed adaptations in the proteome, which does not rely solely on increases or decreases of a same subset of modified proteins.

The specific detection of cysteine-modified IDH under nitrogen starvation is particularly noteworthy, given that this enzyme is located at the crossroad between the nitrogen and carbon metabolisms in cyanobacteria. Isocitrate dehydrogenase provides a carbon backbone (2-oxoglutarate) for the reaction catalysed by glutamate synthase, allowing the net production of two glutamate molecules from glutamine and 2-oxoglutarate. The regulation of this enzyme by cysteine modification could confer a control system at a pivotal place in the cyanobacterial metabolism. It is worth noting that our team has previously described the inactivation of both IDH and GS by oxidative mechanisms (Gómez-Baena *et al.*, 2001; 2006).

Besides the list of proteins specifically detected oxidized under control or N-starved conditions, we observed a number of reversibly oxidized peptides whose levels were either increased or reduced upon N limitation. These are presented in Table 2 and will be discussed in the following paragraphs.

Nitrogen metabolism

The oxidized form of the urea-binding subunit of the putative urea ABC transporter (encoded by the gene *urtD*) in *Prochlorococcus* was found to increase more than fivefold with respect to the control. The urea transporter operon has been identified and characterized in *Synechocystis* PCC 6803 and *Anabaena* PCC 7120 by Valladares and coworkers (Valladares *et al.*, 2002). This protein was also identified as a thioredoxin target in *Synechocystis* sp. PCC 6803 by Lindahl and coworkers (Mata-Cabana *et al.*,



Fig. 3. Changes in isocitrate dehydrogenase concentration in *Prochlorococcus* SS120 cells subjected to nitrogen starvation. Cell extracts were subjected to Western blotting, and isocitrate dehydrogenase detected by using antibodies raised against isocitrate dehydrogenase from *Synechocystis* sp. PCC 6803 (Muro-Pastor *et al.*, 1996). Three independent cultures were divided in two aliquots each, and subjected to control conditions (lanes 1, 2 and 3) or to nitrogen starvation for 24 h (lanes 1-N, 2-N and 3-N).

2007). Interestingly, two of the proteins that comprise the urea transporter in *Prochlorococcus*, *urtA* and *urtB* were found to be among the most upregulated genes in *Prochlorococcus* MED4 and MIT9313 cultures subjected to nitrogen starvation, in studies on global gene expression (Tolonen *et al.*, 2006); but *urtD* was not found to show significant changes. This suggests that changes in regulation induced by nitrogen starvation include both the transcriptional stage [controlled by the master nitrogen regulator, NtcA (Luque and Forchhammer, 2008)] and the post-translational stage, here exemplified by reversible modifications of cysteine residues. Redox modulation of constitutively expressed UrtD to rapidly control its activity (and metabolic flow derived from urea uptake) would complement other regulatory mechanisms of the same process based on transcriptional control of *urtA/urtB* expression.

L-asparaginase activity has been shown to be subjected to non-essential activation by thiolic compounds in *Erwinia carotovora* (Warangkar and Khobragade, 2010). This enzyme catalyses the hydrolysis of asparagine to aspartic acid, removing an amine group. Since our method detects specifically the proteins with reversibly oxidized cysteins, and taking into account that the concentration of the oxidized form was threefold lower in the N-starved cells, it might suggest that the enzyme has become inactive due to N limitation. L-asparaginase was the only protein that had a significantly higher level of the reversibly oxidized peptide in controls in comparison to N-starved cells.

Interestingly, among the detected proteins, there were two enzymes involved in homoserine metabolism (homoserine dehydrogenase and O-acetyl homoserine sulphydrylase). This pathway is involved in the metabolism of threonine and isoleucine and homoserine dehydrogenase has been shown to be inactivated in the presence of cysteine (Hama *et al.*, 1991).

Bifunctional 3,4-dihydroxy-2-butanone 4-phosphate synthase was also detected and is involved in riboflavin biosynthesis from ribulose 5-phosphate. Its yeast orthologue (Rib3p) also has an unrelated and unknown function in mitochondrial respiration. We have recently identified Rib3p in yeast as being a substrate for glutaredoxin 2 and subject to various redox interconversions including glutathionylation (McDonagh *et al.*, 2011b).

Photosynthetic metabolism

Given the importance of photosynthesis in an organism which is universally recognized as one of the main primary producers on Earth, it was somehow expectable to find that several of the proteins affected by nitrogen starvation are directly related to photosynthesis.

Protein D1 is one of the most studied proteins in photosystem II, including cyanobacteria. Cross-linking studies have shown that cysteine residues in elongating D1 nascent chains are located close to the cysteine residues in the D2 protein, suggesting that the formation of a disulfide bond might be important for the stability of D1/D2 early assembled intermediates (Zhang *et al.*, 2000). If this hypothesis holds true for *Prochlorococcus*, it might be possible that the intermediates are blocked in such an early stage due to metabolic limitations preventing the biosynthesis of the mature forms of both proteins. However, the lack of D2 proteins in the list of detected proteins under our experimental conditions might suggest that the cysteine residues from D1 might be also involved in the linking to other proteins or the corresponding cysteine-containing peptide in D2 may not be amenable to mass spectrometry analysis.

Flavodoxin is a photosystem I-electron acceptor, produced in many cyanobacteria as a replacement for ferredoxin under conditions of iron stress (Bottin and Lagoutte, 1992). The flavodoxin sequence in *Synechocystis* PCC 6803 contains four conserved cysteine residues belonging to the iron-sulfur cluster, and two additional cysteine residues which are conserved. The latter two were the residues detected in this work as reversibly modified, indicating that they might be involved in the regulation of this protein under stress conditions. It is worth noting that we detected this protein oxidized both in the presence and in the absence of nitrogen, although this was not under iron limiting conditions.

Coproporphyrinogen II oxidase converts coproporphyrinogen III to protoporphyrinogen IX, the sixth step in the porphyrin biosynthetic route. *Synechocystis* PCC 6803 has two different forms of this enzyme (SII1876 and SII1917) with a common, conserved cysteine motif (Goto *et al.*, 2010), which had been previously described in the crystal structure of this protein in *Escherichia coli* (Layer *et al.*, 2003). The conserved cysteine motif invariably covers a central 4Fe/4S cluster and S-adenosyl-methionine-binding site in a loop, shielding them while providing a route of electron transfer from an electron donor docked to the exterior of the molecule.

The PsaC protein (subunit VII of photosystem I) has two cysteine residues (positions 14 and 51) ligating the Fa and Fb iron-sulfur clusters to photosystem I. By replacing these residues, it has been demonstrated that the presence of [4S-4Fe] clusters in PsaC is required for the assembly of photosystem I cores *in vivo* (Jung *et al.*, 1997). Under conditions of nitrogen limitation, a general response in cyanobacteria is the degradation of the phycobilisomes, which contain a significant amount of nitrogen. *Prochlorococcus* lacks phycobilisomes, but it might be possible that other components of photosystems, as PsaC, be degraded in order to provide nitrogen for more

essential functions under conditions of strong nitrogen limitation, leading to the oxidations of the cysteines.

Glycolytic enzymes

Triose phosphate isomerase, as previously reported, is a dimer containing two cysteine residues/subunit, Cys16 and Cys127, which could form a disulfide upon conformational changes given their close proximity in the molecule and are sensitive to oxidation (McDonagh *et al.*, 2009).

Fructose 1,6-bisphosphatase from *Synechocystis* PCC 6803 is regulated by thioredoxin A and B, hence subjected to redox regulation by reversible modification of cysteine. The reversibly oxidized peptides of both enzymes had double concentration in N-starved cells with respect to control samples.

Phosphorus metabolism

Inorganic pyrophosphatase has been characterized by heterologous expression of two paralogous genes (*ppa1* and *ppa2*) from *Prochlorococcus* MED4 in *Synechocystis* PCC 6803 (Gómez-García and Serrano, 2002). This study showed that the typical cyanobacterial gene (*ppa1*) encodes an inactive enzyme, while the gene *ppa2* (presumably from proteobacterial origin) encodes an active form of pyrophosphatase. Interestingly, all the cyanobacterial *ppa2* genes show three conserved cysteine residues (not shown); two of these cysteines (cys71 and 163) were detected as reversibly modified in this study. This fact had not been noticed thus far, and strongly suggest that these cysteines are important for the function of pyrophosphatase, even if they are not directly related to the catalytic centre, according to reports (Gómez-García and Serrano, 2002).

Other proteins

The SecA gene product likely mediates protein translocation across both the cytoplasmic and thylakoid membranes in cyanobacterial cells: this gene has been described in the cyanobacterium *Synechococcus* PCC 7942 (Nakai *et al.*, 1994). It appears mainly as a soluble homodimer in the cytosol, but a significant amount of SecA was also detected in the membrane after fractionation of cell extracts. In good agreement, SecA was also detected in proteomic studies on membrane proteins interacting with thioredoxin (Mata-Cabana *et al.*, 2007), indicating that the reversibly modified cysteine residues could be involved in this interaction. As suggested by Lindahl and coworkers, and reinforced by the present study, the presence in the membrane of several transporters (such as the urea-binding subunit of the ABC trans-

porter for urea, or SecA) with sensitive cysteine residues might suggest that their functions are regulated (presumably, in the case of N starvation, in order to activate them) by the redox conditions of the cell. Global studies of gene expression showed that N limitation in *Prochlorococcus* provokes the increased expression of a number of genes encoding transporters, such as the ammonium and urea transporters (in strains MED4 and MIT9313), the cyanate transporter in MED4, and the nitrite permease in MIT9313 (Tolonen *et al.*, 2006).

The chaperonin GroEL has been previously reported as a thioredoxin target (Lindahl and Florencio, 2003; Mata-Cabana *et al.*, 2007), and consequently the changes in the redox status of the cysteines interacting with thioredoxin, derived from N limitation, suggest that thioredoxin is mediating the physiological response to N stress in *Prochlorococcus* SS120.

The beta subunit of ATP synthase is encoded by the *atpD* gene. It has been recently demonstrated that chloroplast F(1)-ATPase is subject to redox regulation (Kim *et al.*, 2011), whereby ATP hydrolysis activity is regulated by formation and reduction of the disulfide bond located on the gamma subunit. A similar capability of regulation, by means of reversible modification of cysteine 227, might happen in the beta subunit of ATP synthase, detected in the present study.

The RNA-binding protein RNP-1 contains a RNA-recognizing motif highly conserved in cyanobacteria (Maruyama *et al.*, 1999). Interestingly, an alignment of cyanobacterial RNP-1 proteins shows a single cysteine residue specifically conserved in most marine strains (highlighted with a box in Fig. 4), while other amino acid residues appear at this position in other large group of strains. This might point out to a regulatory feature specific of the majority of the marine cluster, although this hypothesis should be experimentally tested.

Comparison to previous proteomic studies on cyanobacteria subjected to N starvation

The only proteome study which has described the effect of N starvation in cyanobacteria to our knowledge was carried out by Wegener and coworkers (Wegener *et al.*, 2010), and included a quantification of the model cyanobacterium *Synechocystis* PCC 6803 proteome under a wide series of environmental perturbations, including nitrogen depletion, nitrate repletion and ammonium repletion. It is worth noting that the study on *Synechocystis* was focused on total quantification of proteins, while in the present study we have specifically detected proteins subjected to reversible cysteine modification. In this regard the redox sensitive proteins we have detected show a high overlap with those identified as thioredoxin targets in *Synechocystis* sp. (Lindahl and Kieselbach, 2009).

AS9601	gi 123967549	-----MSIFVGNLPPFRAEREDVIQLFAPFGEVLNC
MIT9301	gi 126695350	-----MSIFVGNLPPFRAEREDAIQLFAPFGEVLNC
MIT9312	gi 78778398	-----MSIFVGNLPPFRAEREDVLQLFAPFGEVLNC
MIT9215	gi 157412351	-----MSIFVGNLPPFRAEREDVIQLFAPFGEVVC
MIT9211	gi 159902554	-----MSIFVGNLPPFRAEQEDVIQLFAPFGEVANC
MIT9515	gi 123965247	-----MSIFVGNLPPFRAEREDILELFTPYGEVMNC
NATL1A	gi 124024725	-----MSIFVGNLPPFRAEQEDVMELFSPPFGEVSN
NATL2A	gi 72383177	-----MSIFVGNLPPFRAEQEDVMELFSPPFGEVSN
MED4	gi 33860573	-----MYMNYLISSFYLGYYFKVEFQVSIFVGNLPPFRAEREDILELFTPYGEVMNC
MIT9313	gi 33862287	-----MSIFVGNLPPFRAEQEDVIELFAPFGEVANC
MIT9303	gi 124021728	-----MSIFVGNLPPFRAEQEDVIELFAPFGEVANC
RS9916	gi 116074289	-----MSIFVGNLPPFRAEQEDIIELFSAHGEVTNC
CC9605	gi 78211572	-----MSIFVGNLPPFRAEQEDVIELFAQFGEVTNC
RCC307	gi 148241113	-----MSIFVGNLPPFRAEQEDIIELFASFGEVANC
WH8109	gi 260435624	-----MSIFVGNLPPFRAEQEDVIELFAQFGEVTNC
CC9902	gi 78183598	-----MSIFVGNLPPFRAEQEDVTELFQAQFGEVVC
PCC7001	gi 254431254	-----MSIFVGNLPPFRAEQEDVAELFAPFGEVVC
BL107	gi 116071793	MEHHFVSELWPDGNVPGPFPPFTRSTQNVSIFVGNLPPFRAEQEDVTELFQAQFGEVVC
CB0101	gi 318042980	-----MSIFVGNLPPFRAEQEDVAELFAPFGEVTNC
CB0205	gi 317968520	-----MSIFVGNLPPFRAEQEDVAELFAPFGEVANC
CC9311	gi 113955034	-----MSIFVGNLPPFRAEQEDIIELFSTYGEVTNC
WH7805	gi 88809350	-----MSIFVGNLPPFRAEQEDVIELFAAFGEVTNC
WH7803	gi 148238350	-----MSIFVGNLPPFRAEQEDVIELFAAFGEVTNC
MIT9202	gi 254526608	-----MFAPFGEVVC
RS9917	gi 87123335	-----MSIFVGNLPPFRAEQEDVIELFAAYGEVTNC
PCC7001	gi 254432360	-----MTIYVGNLSFDAEVDVQHLFSEYGDVRC
MIT9303	gi 124022676	-----MTIYIGNLSFQAEQEDLLDLFSQYGEVKTC
MIT9313	gi 33863352	-----MTIYIGNLSFQAEQEDLLDLFSQYGEVKTC
PCC7001	gi 254432068	-----MTIYVGNLSFQAEQEDLLDLFGQYGEVRC
WH5701	gi 87301157	-----MTIYVGNLSFDAEQEDLRLDLFSQYGEVRC
RS9916	gi 116073022	-----MTIYIGNLSFQAEQEDLLDLFSQYGEVKSA
RS9917	gi 87124729	-----MTIYIGNLSFQAEQEDLLDLFNQYGEVKSA
CC9311	gi 113953042	-----MTIYIGNLSFQAEQEDLLDLFGQYGEVKSA
WH7805	gi 88808896	-----MTIYIGNLSFQAEQEDLLDLFSQYGEVKSA
WH7803	gi 148239850	-----MTIYIGNLSFQAEQEDLLDLFSQYGEVKSA
PCC7421	gi 37520960	-----MTLFGVGNLPPFSATEQIEVEAFTEYGEVKSV

Fig. 4. Clustal W alignment of cyanobacterial RNP-1 proteins. The aligned sequences correspond to cyanobacterial *mp1* genes available at Microbial Integrated Genomes, retrieved on 7 July, 2011. The cyanobacterial strain and gene identifier are provided for each protein. The conserved cysteine is highlighted.

Among the proteins which showed significant changes, discussed in this paper, we found that some of them were not detected by Wegener and coworkers; these included subunit II of photosystem I, the RNA recognition motif-containing protein, O-acetyl homoserine sulphydrilase, triosephosphate isomerase and hydroxymyristoyl-ACP dehydratase. In other cases, such as fructose 1,6-bisphosphatase II, flavodoxin, phosphoribosylformylglycinamide synthase, coproporphyrinogen III oxidase, homoserine dehydrogenase, ATP synthase subunit beta, L-asparaginase, reaction centre D1 of photosystem II and protein translocase SecA, the proteins were not detected under either N depletion or ammonium repletion. However, many of these have been identified as potential disulfide proteome Trx targets in *Synechocystis* sp. (Lindahl and Kieselbach, 2009). Taking into account that *Prochlorococcus* SS120 (as all other cultured strains thus far) is incapable of nitrate utilization (López-Lozano *et al.*, 2002; García-Fernández *et al.*, 2004), we focused our comparison to the N depletion vs ammonium repletion

conditions described by Wegener and coworkers. Hereafter we will summarize the comparisons for the three cases for which data were provided (expressed as log₂ ratios in the supplemental table 6 of the manuscript by Wegener and coworkers) under both N depletion and ammonium repletion:

GroEl from *Prochlorococcus* SS120 corresponds to GroEl1 from *Synechocystis* PCC 6803. This protein was much less abundant under N starvation than on ammonium (−0.845 vs 0.692). The difference was less marked for the urea transporter subunit UrtA (1 vs 1.273). Finally, for the RNA polymerase subunit beta (RpoB), again there was a clear decrease under N starvation (−1.17) vs ammonium repletion (1.946).

Since reversible cysteine modification is often involved in the regulation of the protein function, it is possible that some of the proteins detected in our study will show no significant change in their concentration under N limitation such as GS (Fig. 2). The redox modification can produce the required effects without need to modify their

concentrations, either by new protein synthesis or by degradation.

On the other hand, in the three proteins which allowed direct comparison (GroEI, UrtA and RpoB), it seems that the general effect of nitrogen starvation was to decrease their concentration. However, the actual effect of nitrogen on the regulation of the metabolism might be more complex; for instance, microarray studies on *Prochlorococcus* showed that the expression of *urtA* was upregulated (41.36-fold in the strain MED4 and 9.45-fold in the strain MIT9313), in contrast to the lower concentration of UrtA in *Synechocystis* PCC 6803 under N starvation. This apparent contradiction might be explained by a very strong turnover of UrtA under N limitation or by a very different physiological response in *Prochlorococcus* vs *Synechocystis*. Our results demonstrate that besides possible changes of *urtA* expression/UrtA concentration in *Prochlorococcus* SS120, this subunit of the transporter seems to be subjected to regulation by cysteine modification, pointing out to an important role of this ABC transporter under nitrogen limitation. This is in good agreement to the general response to stress, proposed by Wegener and coworkers involving the activation of urea utilization as a source of nitrogen (Wegener *et al.*, 2010).

Concluding remarks

Nitrogen starvation had a wide range of effects on the redox proteome of *Prochlorococcus*. Key enzymes from the nitrogen metabolism were modified, but at the same time we observed changes in transporters and transaminases, suggesting that *Prochlorococcus* was adapting by activation of the uptake of nitrogen and by transferring nitrogen to essential proteins.

On the other hand, we could detect a strong effect on photosynthetic structural proteins and also enzymes involved in chlorophyll biosynthesis or in electron transfer.

In addition, we observed that nitrogen limitation affected phosphorus metabolism, the status of ATP synthase, and enzymes involved in the nucleic acids metabolism. This suggests that the lack of nitrogen affects almost all metabolic pathways, far beyond the proteins directly involved in nitrogen metabolism. This is in good agreement to the observation by Tolonen and coworkers (Tolonen *et al.*, 2006), who found that many of the genes whose expression was affected by N starvation did not belong to the large group of genes controlled by the master N regulator, NtcA.

In conclusion, our results demonstrate that beyond the transcriptional changes elicited by nitrogen starvation, *Prochlorococcus* also responds with extensive post-translational redox modifications, a reflection of the importance of nitrogen in cyanobacterial metabolism.

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Supporting information

Additional Supporting Information may be found in the online version of this article:

Table S1. Three independent cultures ($n = 3$) for both control and N-starved *Prochlorococcus* cells were analysed by MS/MS and an excel table containing all proteins that were identified from at least two independent cultures is presented. The Sequest score of proteins, the % coverage and number of unique peptides for each protein from each independent culture and from each strain can be examined.

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